# Drivers of Pain in Daily Life: An EMA-Based Idiographic Network Analysis of Attention, Threat, Flow, and Activity Engagement in Chronic Pain

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**Abstract**

According to the cognitive-affective model of the interruptive function of pain, pain preferentially recruits attention, the strength of this recruitment is governed by the threat value of pain rather than by its intensity alone, and the resulting capture of attention competes with the pursuit of goal-directed activity. To examine these dynamics in daily life, patients with chronic pain ($N = 68$ in the analytic sample) completed a momentary diary eight times per day for two weeks (6,262 completed prompts, median = 96 per person, range 30 to 122). Momentary pain, its threat value (the mean of pain-related fear and catastrophizing), attention to pain, and goal engagement were modelled as within-person temporal networks. Because the series reach the length recommended for individual estimation, person-specific networks formed the primary analysis, with a multilevel vector autoregressive (mlVAR) model as the partial-pooling benchmark and S-GIMME as a confirmatory estimator. The benchmark recovered a coherent within-person threat-attention-pain circuit: a higher than usual threat value predicted a stronger subsequent focus on pain ($b = .125$, 95% CI $[.078, .172]$), higher subsequent pain ($b = .096$, $[.047, .145]$), and pain in turn predicted a stronger subsequent focus on pain ($b = .083$, $[.046, .120]$); attention also predicted higher subsequent pain ($b = .038$, $[.003, .073]$). These average effects summarized wide heterogeneity: the per-person threat-to-pain coupling ranged from $-.24$ to $.86$, was individually reliable in 11 of 66 participants, and was positive in every participant for whom the regularized individual network selected it. Between persons, a baseline threat-value profile (catastrophizing and vigilance) predicted more attention to pain ($r = .39$, $p = .001$) and less goal engagement ($r = -.30$, $p = .01$), and did so more specifically than biomedical severity and neuroticism. Because goal engagement is also the absorption component of flow, the activity items were additionally combined into a momentary flow experience: it was concurrently associated with lower pain ($b = -.094$, $p < .001$) and lower pain interference ($b = -.139$, $p < .001$), showed no lagged association in either direction, and reproduced the benchmark network almost exactly, with the association carried by absorption and perceived skill rather than by the challenge-skill balance the construct is usually defined by. Attention and threat, rather than pain intensity alone, appear to drive the momentary experience of chronic pain, in a way that is directionally shared across patients but strongly person-specific in magnitude.

*Keywords:* chronic pain; attention to pain; threat value of pain; flow; ecological momentary assessment; idiographic networks; multilevel vector autoregression

# 1 Introduction

## 1.1 Pain demands attention

Pain interrupts what a person is doing and demands attention, an adaptive property that becomes a burden when pain persists (Eccleston and Crombez, 1999). In the cognitive-affective model of the interruptive function of pain, the degree to which pain captures attention and disrupts ongoing behaviour is not fixed by pain intensity. It depends on the threat value of the pain, that is, on how far the pain is appraised as dangerous or as signalling harm, and on the characteristics of the activity that the pain interrupts (Eccleston and Crombez, 1999, 2007). Experimental work established that the threat value of somatic information, carried by pain-related fear and catastrophic thinking, enhances the attentional interference produced by pain (Crombez et al., 1998, 1999, 2013). Pain-related fear has been shown to be more disabling than pain itself (Crombez et al., 1999), and attention to pain is a candidate process linking that threat to daily disability (Van Damme et al., 2010).

Whether these laboratory processes operate in the everyday life of patients with chronic pain has been examined with experience-sampling (diary) methods, which repeatedly sample momentary pain, pain-related appraisals, attention to pain, and ongoing activity in the natural environment (Tennen and Affleck, 1996; Peters et al., 2000). Such methods make it possible to ask whether the threat value of pain and the attention it recruits precede momentary pain in daily life, as the interruptive-function model predicts, and whether the activity that pain interrupts is displaced when pain and attention rise.

## 1.2 From group averages to within-person dynamics

Most evidence on the determinants of pain and disability is between persons. Such designs show whether patients who are more fearful or more vigilant are, on average, more disabled, but they cannot answer the within-person prospective question that the interruptive-function model implies: when a given patient attends to pain more than usual, or appraises it as more threatening than usual, does pain then rise, and does this differ across patients? Between-person and within-person associations need not agree in size or even in sign (Curran and Bauer, 2011; Fisher et al., 2018), and processes that unfold within individuals over time require models specified at that level (Molenaar, 2004; Hamaker and Wichers, 2017; Wright and Woods, 2020). Experience sampling supports this aim by repeatedly measuring experience in daily life (Myin-Germeys et al., 2018; Shiffman et al., 2008; Schwartz and Stone, 1998).

Network models offer a natural language for such data: constructs are nodes, and temporal or contemporaneous associations are edges (Borsboom and Cramer, 2013; Bringmann et al., 2013). Two estimators are widely used. Multilevel vector autoregression (mlVAR; Bringmann et al., 2013; Epskamp et al., 2018b) estimates a group-level network while allowing person-specific deviations through random effects. Group iterative multiple model estimation, including its subgrouping extension S-GIMME (Gates and Molenaar, 2012; Lane et al., 2019), searches for group, subgroup, and individual paths. A practical constraint is series length: stable fully individual networks typically require on the order of 100 observations per person, a bar that many brief experience-sampling protocols do not clear (Beltz et al., 2016). Whether idiographic estimation is defensible therefore depends on the density and duration of the sampling protocol at hand.

### 1.3 The present study

The present study uses a chronic-pain experience-sampling dataset in which patients completed a battery of questionnaires and then, using a palmtop computer, answered a short momentary diary eight times per day for two weeks, together with a brief morning and evening diary. The analysis asks what drives momentary pain in daily life. With a median of 96 completed prompts per person, individual estimation is defensible, unlike in briefer protocols such as a companion study in which about 44 prompts per person were available. Guided by the interruptive-function model, momentary pain is modelled together with its threat value (the appraisal of pain as dangerous, indexed by momentary fear and catastrophizing), attention to pain, and goal engagement (absorption in the ongoing activity that pain interrupts). Because momentary pain, threat, attention, and engagement are mutually entangled and the data are observational, the temporal edges are read as within-person predictive associations rather than as causal effects (Rohrer, 2018). Two research questions guide the analysis.

**RQ1.** How do momentary threat value, attention to pain, and goal engagement relate temporally to momentary pain at the level of the individual patient, and how much do these person-specific dynamics differ across patients?

**RQ2.** Are individual differences in the momentary experience of pain accounted for by a baseline threat-value profile (catastrophizing, vigilance), over and above a biomedical profile (pain severity, duration) and a personality profile (neuroticism)?

## 2 Method

### 2.1 Participants and design

Patients with chronic musculoskeletal pain, recruited through a university pain clinic, first completed a battery of validated questionnaires and then carried a palmtop computer that signalled them eight times per day for two weeks to complete a short momentary diary, together with a brief morning and evening diary. All 68 participants with momentary diary data met the compliance criterion (at least 25 completed prompts) and formed the analytic sample, contributing 6,262 completed momentary prompts (median = 96 per person, mean = 92.1, range = 30 to 122). Sixty-five of the 68 participants completed at least 50 prompts, and most approached or exceeded the series length usually recommended for individual network estimation. Baseline questionnaires were available for 67 participants. Sample characteristics are summarized in Table 1.

### 2.2 Momentary measures

At each prompt, participants rated a set of items on seven-point scales (1 = not at all, 7 = very much). Momentary pain was the pain-intensity item ("I am in pain now"). Attention to pain was the item "I focus on my pain"; a second, reverse-keyed item ("I can ignore the pain now") was combined with it to form a two-item attention composite used in a sensitivity analysis. The threat value of pain was the mean of momentary fear ("I am afraid of the pain now") and catastrophizing ("I feel that my pain is becoming too much for me"), the two appraisals that, in the interruptive-function model, amplify the attentional capture

**Table 1.** Baseline characteristics of the analytic sample.

| Variable | Mean (SD) | Range | n |
|---|---|---|---|
| Age (years) | 46.4 (9.8) | 21 to 65 | 67 |
| Sex: women / men (%) | 80.6 / 19.4 | | 67 |
| Pain duration (months) | 82.8 (121.4) | 2 to 520 | 62 |
| MPI pain severity | 4.3 (0.9) | 2 to 6 | 67 |
| MPI interference | 4.2 (1.0) | 2 to 6 | 67 |
| Pain Disability Index | 42.7 (11.8) | 11 to 68 | 67 |
| PCS catastrophizing (total) | 22.3 (10.2) | 1 to 46 | 66 |
| PVAQ pain vigilance (total) | 43.7 (12.8) | 5 to 66 | 65 |
| HADS anxiety | 7.7 (3.9) | 1 to 20 | 66 |
| HADS depression | 8.7 (3.7) | 1 to 16 | 66 |
| NEO-FFI neuroticism | 30.8 (9.4) | 12 to 54 | 64 |

*Note.* MPI = West Haven-Yale Multidimensional Pain Inventory; PDI = Pain Disability Index; PCS = Pain Catastrophizing Scale; PVAQ = Pain Vigilance and Awareness Questionnaire; HADS = Hospital Anxiety and Depression Scale; NEO-FFI = NEO Five-Factor Inventory.

by pain; the two items co-varied within persons ($r$ = .41) and across persons ($r$ = .68), supporting their combination into a single threat node, and a model in which they were entered separately is reported as a sensitivity analysis. Goal engagement was "I am absorbed in this activity", the central goal-directed activity characteristic in the diary. Two further goal-directed characteristics, efficacy ("I am good at this") and valence ("I like doing this"), were used in an extended model. Together with a challenge item ("this activity is a personal challenge for me"), these activity appraisals also index flow, a state of deep absorption and effortless concentration arising when perceived challenge and perceived skill are both high and matched, and experienced as intrinsically rewarding (Csikszentmihalyi, 1990; Nakamura and Csikszentmihalyi, 2014); their combination is described below. Participants also reported their location, whether they were alone, whether they had taken pain medication, and whether the prompt disturbed them. The exact item wording and the mapping to analytic variables are documented in the study codebook.

## 2.3 Baseline measures

Baseline questionnaires provided the trait profiles for RQ2. The threat-value profile combined the Pain Catastrophizing Scale total (Sullivan et al., 1995) and the Pain Vigilance and Awareness Questionnaire total (Roelofs et al., 2003), the two constructs that capture threat-related appraisal and monitoring. The biomedical profile combined the West Haven-Yale Multidimensional Pain Inventory pain-severity subscale (Kerns et al., 1985) and pain duration. The personality profile was NEO-FFI neuroticism (Costa and McCrae, 1992), which correlated positively with concurrent distress on the Hospital Anxiety and Depression Scale ($r$ = .73), supporting its scoring. Each profile was standardized.

## 2.4 Statistical modelling

Analyses were implemented in a reproducible pipeline in which statistical models were written in R (R Core Team, 2024) and orchestrated from Python, which handled preprocessing, tables, and visualization. Each modelling step is a separate script, and all reported tables and figures were regenerated from the processed data.

**Preprocessing and time structure.** Momentary items were scored and, where relevant, reverse keyed before analysis. A prompt was coded as completed when the pain item was answered. Temporal predictors were formed from adjacent prompts within the same study day; no lag was carried across the overnight gap. Variables were person-standardized before temporal modelling, so that lag-1 coefficients can be read as standardized within-person effects. Weak stationarity was examined for each person and variable with linear-trend and KPSS tests, which motivated a within-person detrended sensitivity model.

**Variance decomposition.** Null multilevel models with random person intercepts and nested person-day intercepts (Bates et al., 2015) partitioned each momentary measure into between-person, between-day, and within-day variance. The within-day component is the moment-to-moment variance available to the lag-1 networks.

**Primary idiographic networks for RQ1.** Because the series reach the recommended length, fully individual networks were the primary analysis. For each participant a regularized network was estimated with graphicalVAR, which selects a sparse lag-1 temporal network and a contemporaneous network by extended Bayesian information criterion (Epskamp et al., 2018a). To quantify the spread and the precision of the pain-driver couplings, an unregularized within-person model of the pain equation was also fitted for each participant, regressing momentary pain on the previous prompt's pain, threat, attention, and engagement, and its driver-to-pain coefficients were retained with 95% confidence intervals.

**Pooled benchmark and confirmatory estimator.** A multilevel VAR model (Epskamp et al., 2018b) on the four core nodes provided the partial-pooling benchmark. It estimates a directed lag-1 temporal network, an undirected contemporaneous network of same-prompt residual partial correlations, and a between-person network of partial correlations among person means. Random effects were orthogonal. The temporal fixed effects give the average within-person network, and the random-effect standard deviations index between-person heterogeneity. S-GIMME (Lane et al., 2019) was run as a confirmatory estimator to check whether a second data-driven method recovered the same backbone.

**RQ2 analyses.** Two complementary analyses were used. First, each participant's mean momentary attention to pain and mean goal engagement were related to the three baseline profiles with zero-order and partial (controlling mean momentary pain) correlations and a hierarchical regression. Second, a cross-level moderation tested whether a baseline profile shapes the *within-person* coupling strength: for each of five pre-specified couplings, a multilevel model estimated the coupling as a random slope and its interaction with each standardized profile. This one-step test replaces the underpowered correlation of two-step per-person coefficients with baseline traits.

**Flow as a composite construct.** Goal engagement, one of the four core nodes, is the absorption component of flow, so the construct is partly present in the model already. Following the standard experience-sampling operationalization (Massimini and Carli, 1988; Moneta and Csikszentmihalyi, 1996), flow was represented by two structurally distinct parts rather than by a single sum, because summing the constituents conflates the antecedent condition with the outcome experience. The condition was expressed as balance and elevation, and the experience as the mean of absorption and enjoyment:

$$\text{balance} = -|C - S|, \qquad \text{elevation} = \tfrac{1}{2}(C + S), \qquad \text{flow experience} = \tfrac{1}{2}(A + E), \tag{1}$$

with $C$ challenge, $S$ skill, $A$ absorption, and $E$ enjoyment. The elevation term is required because balance alone would score a moment low on both challenge and skill, that is apathy, as perfectly balanced. Three further decisions matter for interpretation. First, the metric: person-mean centering was primary (Curran and Bauer, 2011), because within-person standardization sets every participant's mean to zero and therefore assigns above-average moments to everyone, including participants who never reach flow in absolute terms; the raw scale, grand-mean $z$, and within-person $z$ were run as sensitivity analyses (Figure S11, Table S13). Second, because momentary challenge co-varied strongly with self-reported physical activation ($r = .57$ within persons), all flow models were also estimated with activation as a covariate. Third, since composite measures can be driven by a single constituent and thereby obscure rather than summarize (VanderWeele, 2022), absorption, enjoyment, challenge, and skill were entered separately as well as jointly, and the benchmark network was re-estimated with the flow composite in place of the single absorption node. The four items were never summed into one score. This is a secondary, analyst-constructed measure: the original study analysed these items separately (Viane et al., 2004).

**Robustness and diagnostics.** The benchmark temporal network was re-estimated after within-person detrending, with the two-item attention composite, with a stricter compliance threshold (at least 70 prompts), and in a six-node extension adding efficacy and valence. Sampling stability was assessed with leave-one-participant-out refitting and a person-level cluster bootstrap (200 resamples). Residual diagnostics examined residual shape, residual-fitted associations, and within-person lag-1 residual autocorrelation. Daily functioning was examined by relating each day's momentary experience to the evening reports of goal success and pain interference in multilevel models.

### 2.5 Literature-derived hypotheses

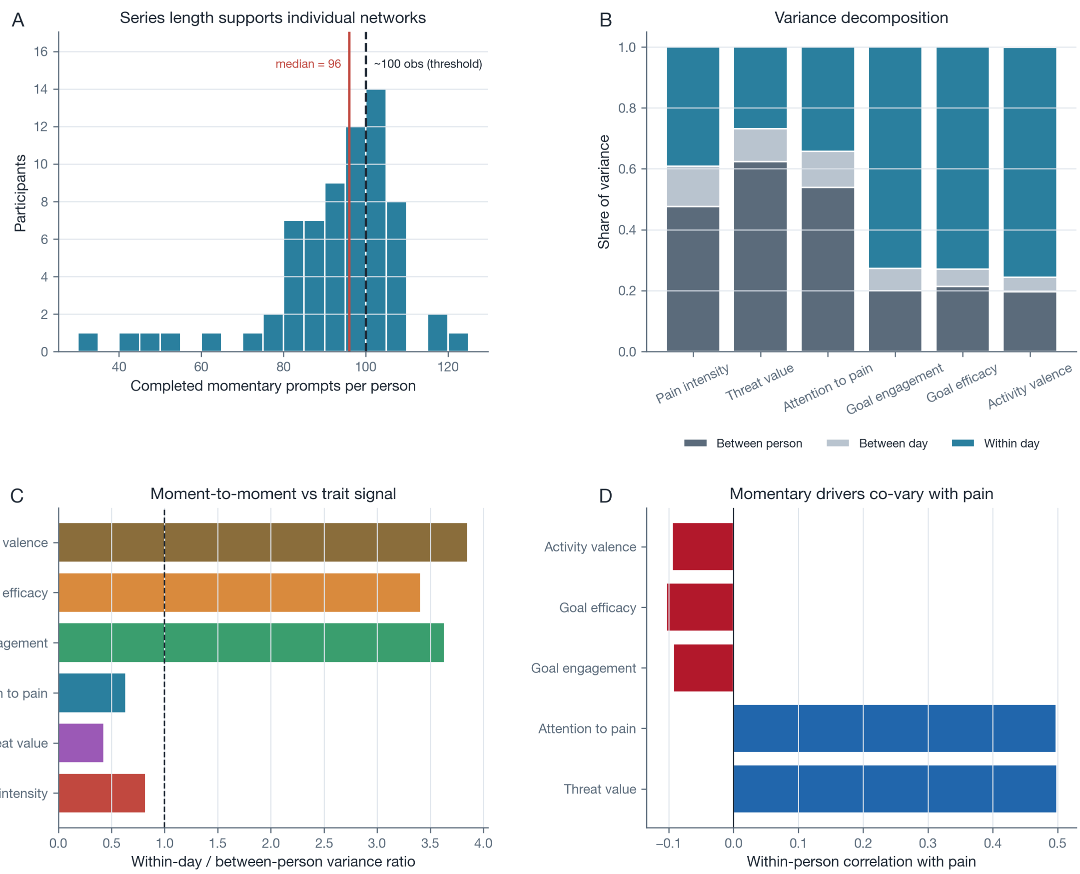
For RQ1, four hypotheses were specified. H1 predicted positive temporal persistence for pain, threat, attention, and engagement. H2 predicted a positive within-person threat-attention-pain circuit: a higher than usual threat value and more attention to pain would be followed by higher pain, and higher pain would be followed by more attention to pain. H3 predicted that higher pain would be followed by less goal engagement, the interruption of goal pursuit. H4 predicted substantial person-specific heterogeneity in these couplings. For RQ2, H5 predicted that the baseline threat-value profile would explain individual differences in the momentary experience of pain better than the biomedical or personality profiles.

## 3 Results

### 3.1 Compliance, within-person variance, and feasibility

Compliance was good for a two-week eight-per-day protocol, with a modest decline across days (Figure S1). The variance decomposition showed that all momentary measures carried substantial within-person signal (Figure 1, Table 2). Momentary pain was split about evenly between and within persons (ICC = .49; within-day share = .39), attention to pain behaved similarly (ICC = .55; within-day = .34), the threat value of pain was more trait-like (ICC = .64; within-day = .27), and goal engagement was overwhelmingly momentary (ICC = .21; within-day = .73). The completed-prompt distribution clustered near the recommended series length for individual estimation (Figure 1, Panel A), which is the basis for foregrounding person-specific networks. Within persons, momentary pain co-varied positively with threat ($r$ = .48) and attention ($r$ = .48), and weakly negatively with engagement, efficacy, and valence (all $r$ between −.08 and −.10; Figure 1, Panel D).

**Figure 1.** Series length, within-person variance, and pain-driver correlations.




*Note.* Panel A shows the distribution of completed prompts per person, with the median and the approximate idiographic series-length threshold marked. Panel B partitions each measure into between-person, between-day, and within-day variance. Panel C shows the within-day to between-person variance ratio. Panel D shows within-person correlations of each driver with pain.

## 3.2 RQ1: A within-person threat-attention-pain circuit

The pooled benchmark provides the average within-person structure against which the person-specific estimates are read (Figure 2, Table 3). All four states were positively self-persistent (pain $b$ = .344, threat $b$ = .297, attention $b$ = .188, engagement $b$ = .149; all $p$ < .001), supporting H1. A coherent threat-attention-pain circuit then emerged, supporting H2. The strongest cross-lagged effect was from threat to attention: a higher than usual threat value predicted a stronger subsequent focus on pain ($b$ = .125, $SE$ = .024, $p$ < .001), the momentary counterpart of the model's claim that threat amplifies the attentional

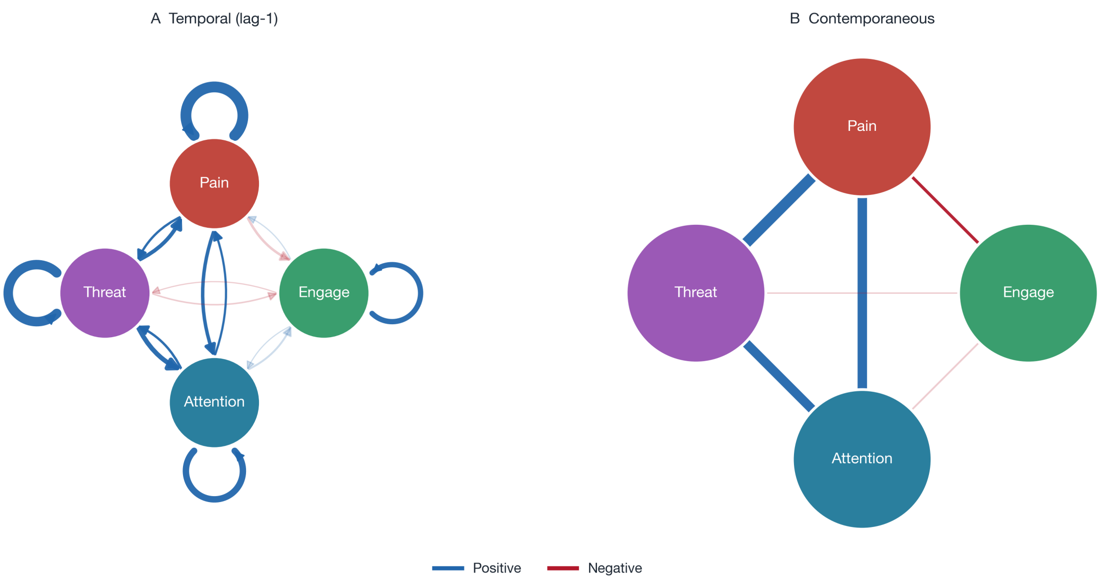
**Table 2.** Momentary measures: descriptive statistics and variance partition.

| Measure | *n* obs | Mean | SD | % miss. | ICC | Within-day var. |
|---|---|---|---|---|---|---|
| Pain intensity | 6262 | 4.53 | 1.53 | 0.0 | 0.48 | 0.39 |
| Threat value | 6222 | 2.24 | 1.38 | 0.6 | 0.64 | 0.27 |
| Attention to pain | 6203 | 2.89 | 1.58 | 0.9 | 0.55 | 0.34 |
| Goal engagement | 6143 | 4.0 | 1.9 | 1.9 | 0.2 | 0.73 |
| Goal efficacy | 6108 | 4.79 | 1.55 | 2.5 | 0.22 | 0.73 |
| Activity valence | 6136 | 4.9 | 1.58 | 2.0 | 0.2 | 0.76 |

*Note.* ICC = intraclass correlation (between-person share). Within-day var. is the moment-to-moment variance share available to the lag-1 network.

capture by pain. Threat also predicted higher subsequent pain ($b$ = .096, $SE$ = .025, $p$ < .001), pain predicted a stronger subsequent focus on pain ($b$ = .083, $SE$ = .019, $p$ < .001), closing a bidirectional pain-attention loop, and attention predicted higher subsequent pain ($b$ = .038, $SE$ = .018, $p$ = .039). The interruption of goal engagement by pain was in the predicted direction, reached only a trend in the pooled Wald test ($b$ = −.048, $SE$ = .026, $p$ = .065), but was supported by the person-level bootstrap (95% CI [−.100, −.003]; see robustness below), giving qualified support to H3. The contemporaneous network was dominated by the same-prompt threat-pain, attention-pain, and threat-attention associations (partial $r$ = .33, .28, and .27, all $p$ < .001), and the between-person network by the trait threat-attention and threat-pain associations (Table S1).

**Figure 2.** Pooled benchmark momentary networks from multilevel VAR.




*Note.* Panel A shows the directed temporal (lag-1) network within day. Panel B shows the contemporaneous network at the same prompt. Blue edges are positive, red edges are negative, edge width scales with strength, and faded edges are not statistically significant. The between-person network is reported in Table S1.

## 3.3 Individual heterogeneity in the pain-driver couplings

The average circuit summarizes wide person-to-person variation, supporting H4 (Figure 3). The unregularized per-person threat-to-pain coefficient ranged from −.24 to .86 (mean = .082), the attention-to-pain coefficient from −.31 to .73 (mean = .068), and the engagement-to-pain coefficient from −.53 to .31

**Table 3.** Pooled temporal (lag-1) network: multilevel VAR fixed effects (benchmark).

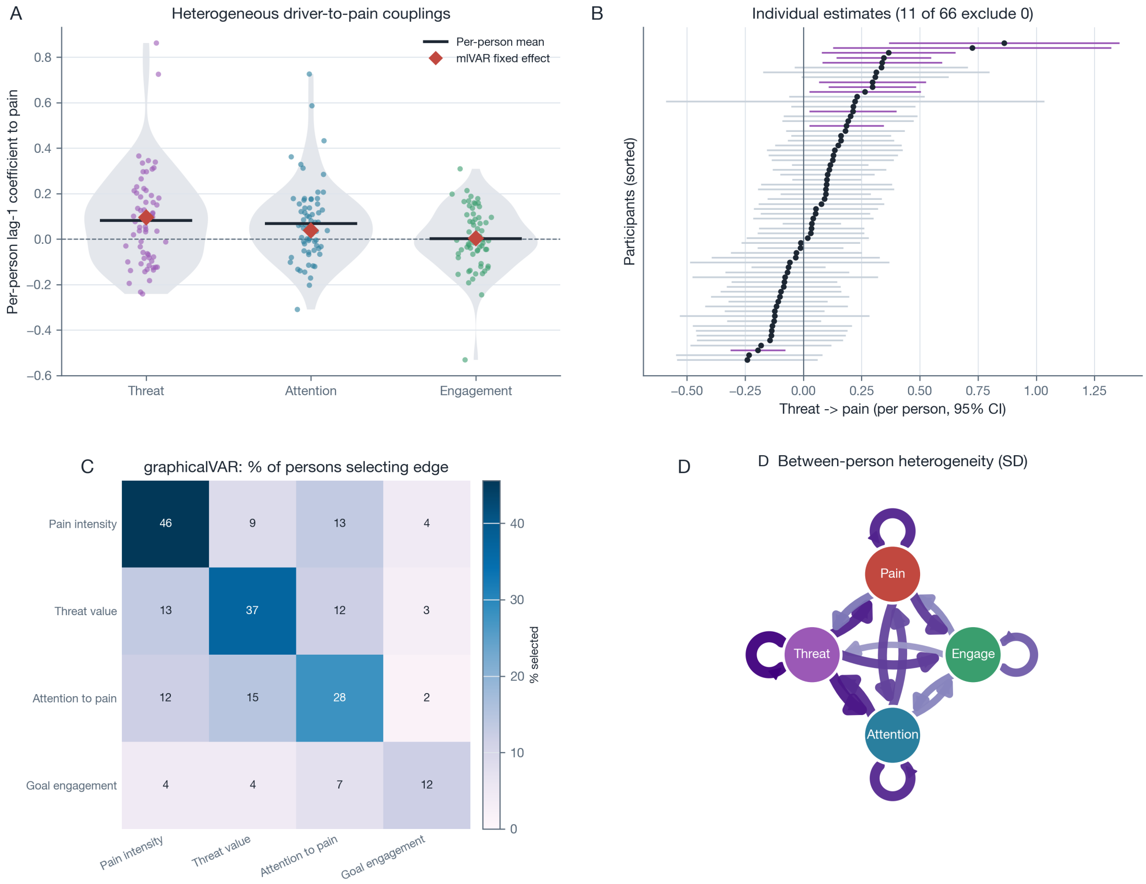
| Effect (lag-1) | Estimate | SE | 95% CI | $p$ | Random SD |
|---|---|---|---|---|---|
| Pain intensity → Attention to pain | 0.083 | 0.019 | [0.05, 0.12] | <.001 | 0.07 |
| Threat value → Pain intensity | 0.096 | 0.025 | [0.05, 0.15] | <.001 | 0.118 |
| Threat value → Attention to pain | 0.125 | 0.024 | [0.08, 0.17] | <.001 | 0.101 |
| Pain intensity → Threat value | 0.043 | 0.013 | [0.02, 0.07] | .001 | 0.0 |
| Attention to pain → Threat value | 0.045 | 0.017 | [0.01, 0.08] | .009 | 0.077 |
| Attention to pain → Pain intensity | 0.038 | 0.018 | [0.00, 0.07] | .039 | 0.067 |
| Pain intensity → Goal engagement | -0.048 | 0.026 | [-0.10, 0.00] | .065 | 0.102 |
| Attention to pain → Goal engagement | 0.028 | 0.023 | [-0.02, 0.07] | .219 | 0.042 |
| Threat value → Goal engagement | -0.014 | 0.03 | [-0.07, 0.04] | .631 | 0.106 |
| Goal engagement → Pain intensity | 0.003 | 0.012 | [-0.02, 0.03] | .809 | 0.049 |
| Goal engagement → Threat value | -0.001 | 0.009 | [-0.02, 0.02] | .886 | 0.0 |
| Goal engagement → Attention to pain | 0.0 | 0.012 | [-0.02, 0.02] | .996 | 0.052 |
| Pain intensity → Pain intensity | 0.344 | 0.025 | [0.29, 0.39] | <.001 | 0.15 |
| Threat value → Threat value | 0.297 | 0.027 | [0.24, 0.35] | <.001 | 0.163 |
| Attention to pain → Attention to pain | 0.188 | 0.025 | [0.14, 0.24] | <.001 | 0.149 |
| Goal engagement → Goal engagement | 0.149 | 0.021 | [0.11, 0.19] | <.001 | 0.125 |

*Note.* Standardized within-person coefficients from the mlVAR benchmark. Cross-lagged effects are listed above autoregressive effects. Random SD is the between-person standard deviation of the effect, the pooled-model index of heterogeneity.

(mean = .001; Figure 3, Panel A). The per-person means matched the pooled fixed effects, but the spread was large and, for engagement, centred on zero with substantial variation in both directions. Individual estimates of the threat-to-pain coupling were individually reliable in 11 of 66 participants, more than expected by chance and all in the positive direction (Figure 3, Panel B). Regularized individual networks were sparse but interpretable: the median number of selected temporal edges was one, edge selection was dominated by the autoregressions, and among the cross-variable edges the threat-to-pain and attention-to-pain edges were the most frequently selected pain-driver edges and were positive whenever selected (Figure 3, Panel C, Figure S4, Table S5). The between-person heterogeneity was structured rather than uniform, with the largest between-person standard deviations on the threat-involving and attention-involving paths (Figure 3, Panel D). S-GIMME, run as a confirmatory estimator, recovered the autoregressive backbone and a shared threat-attention-pain cluster (Figure S5, Table S4). Adding efficacy and valence in a six-node extension left the core four-node structure essentially unchanged (Figure S6, Table S6).

**Flow as a composite of the activity appraisals.** Because goal engagement is the absorption half of flow, the activity items were combined into a momentary flow experience (Equation (1)) and carried through the same models. Within persons, the challenge-skill condition predicted the flow experience strongly, but entirely through elevation ($b = .744$, $SE = .024$, $p < .001$) and not through balance ($b = .006$, $p = .78$); between persons the balance effect was in fact negative ($b = -.111$, $p = .028$), as it was on the raw response scale ($b = -.090$, $p < .001$). An additive model with challenge and skill as separate main effects fitted better than the balance-and-elevation form (AIC 16,880 versus 17,321) and their interaction was not significant ($p = .15$), so in these data the flow condition behaves additively rather than configurationally. Under the conservative absolute criterion 19.3% of moments qualified as flow and four of 68 participants never reached it, whereas the within-person metrics placed no participant at zero, which is why prevalence is reported on the absolute scale only (Table S13). The flow experience was

**Figure 3.** Individual heterogeneity in the pain-driver couplings.

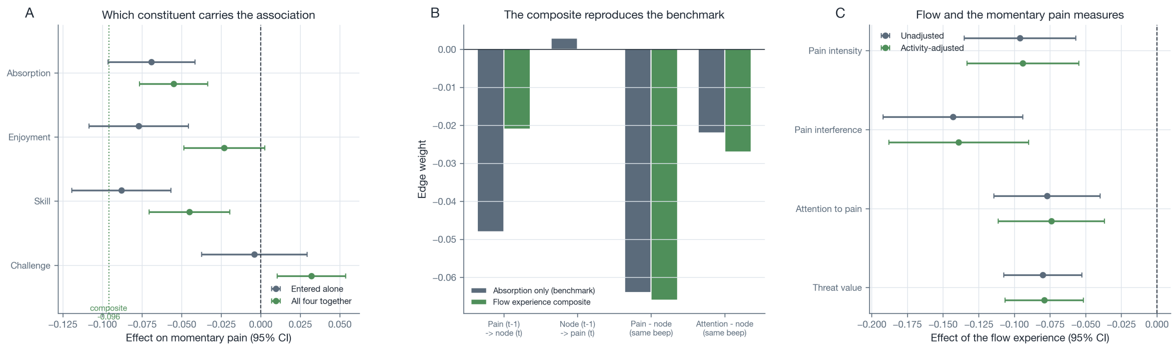



*Note.* Panel A shows the per-person lag-1 coefficients of threat, attention, and engagement predicting pain, with the per-person mean and the mlVAR fixed effect marked. Panel B shows each participant's unregularized threat-to-pain estimate with its 95% confidence interval, sorted; intervals excluding zero are highlighted. Panel C shows the percentage of participants for whom graphicalVAR selected each directed temporal edge. Panel D shows the between-person heterogeneity network, in which edge width and colour intensity scale with the standard deviation of the per-person lag-1 coefficient for that path.

concurrently associated with lower pain ($b = -.094$, $SE = .020$), lower pain interference ($b = -.139$, $SE = .025$), less attention to pain ($b = -.074$, $SE = .019$), and a lower threat value ($b = -.079$, $SE = .014$), all $p < .001$ and all unchanged after adjusting for physical activation (Figure 4, Panel C). No lagged effect reached significance in either direction, while the autoregressive terms remained large (Figure S12, Panel B), so the association is concurrent rather than temporal. It was also person-specific in the same way as the other couplings: the concurrent slope was negative in 73% of participants, reliably so in 14 of 63, and reliably positive in 4 (Figure S12, Panels D to F).

Two results bear on the construct itself. First, replacing the single absorption node by the composite reproduced the benchmark network almost exactly, with edge differences of .003 to .027 and no change of sign or significance (Figure 4, Panel B): the composite adds a label rather than information. Second, the association with pain was not evenly distributed over the constituents (Figure 4, Panel A). Entered alone, absorption ($b = -.069$), enjoyment ($b = -.077$), and skill ($b = -.088$) each tracked lower pain while challenge did not ($b = -.004$, $p = .82$). Entered jointly, absorption ($b = -.055$, $p < .001$) and skill ($b = -.045$, $p < .001$) retained their associations, enjoyment fell to non-significance ($p = .09$), and challenge reversed sign and became positive ($b = .032$, $p = .004$). The composite is therefore carried by two of its four constituents, and one constituent works against the other three, which is the pattern that makes a summed score of such measures misleading (VanderWeele, 2022).

**Figure 4.** Flow as a composite construct.




*Note.* Panel A shows each activity appraisal as a predictor of momentary pain, first entered alone and then with all four entered jointly, with the composite shown as a reference line; all models are person-mean centred and adjusted for physical activation. Panel B compares the pain-relevant edges of the benchmark network, in which the activity slot is occupied by absorption alone, against the same network with the flow composite in that slot. Panel C shows the concurrent association of the flow experience with the four momentary pain measures, unadjusted and adjusted for physical activation. Error bars are 95% confidence intervals.

**Table 4.** Flow states and the momentary experience of pain.

*Panel A.* Challenge-skill condition predicting the flow experience (multilevel, random slopes).

| Term | *b* | SE | *p* | Random SD |
|---|---|---|---|---|
| Balance (within) | 0.006 | 0.020 | .782 | 0.131 |
| Elevation (within) | 0.744 | 0.024 | <.001 | 0.164 |
| Balance (between) | -0.111 | 0.049 | .028 | |
| Elevation (between) | 0.796 | 0.085 | <.001 | |

*Panel B.* Flow experience and the momentary experience of pain, same beep and lag 1.

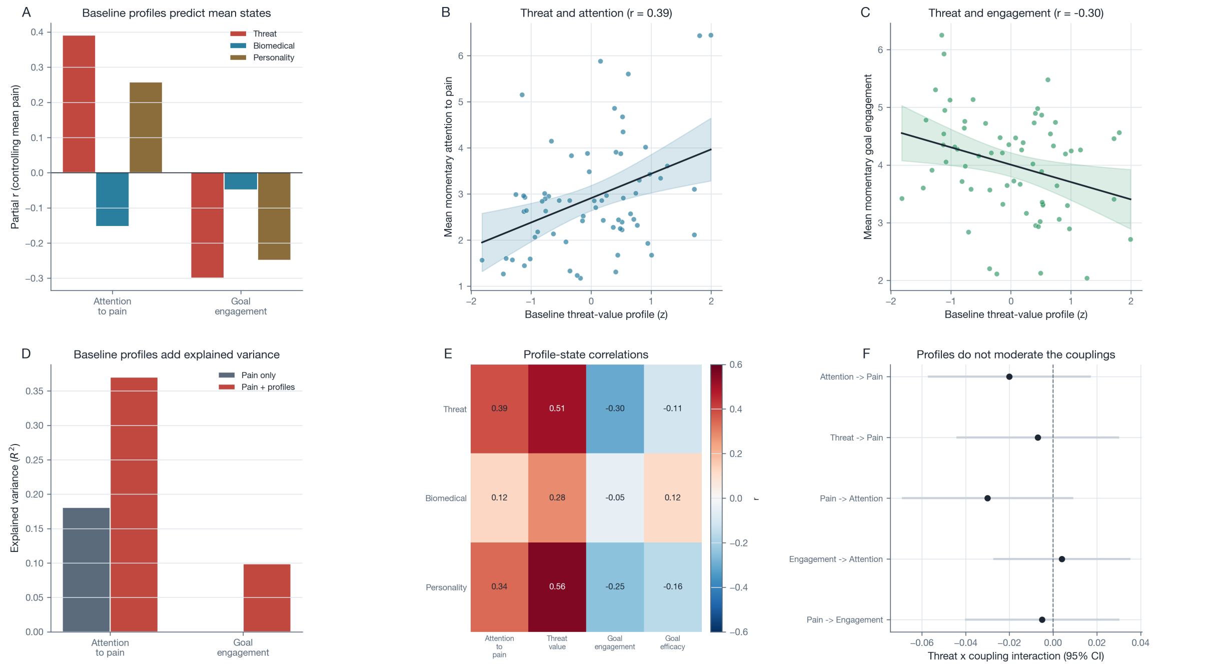
| Outcome | *b* (same beep) | SE | *p* | *b* (lag 1) | SE | *p* |
|---|---|---|---|---|---|---|
| Pain intensity | -0.094 | 0.020 | <.001 | -0.002 | 0.014 | .888 |
| Pain interference | -0.139 | 0.025 | <.001 | 0.001 | 0.014 | .921 |
| Attention to pain | -0.074 | 0.019 | <.001 | -0.012 | 0.016 | .451 |
| Threat value | -0.079 | 0.014 | <.001 | -0.011 | 0.014 | .439 |

*Note.* The flow experience is the mean of absorption and enjoyment; balance is $-|\text{challenge} - \text{skill}|$ and elevation is (challenge + skill)/2, both person-mean centered. Panel B models are adjusted for momentary physical activation; lag-1 models control the outcome's own lag and are estimated on person-standardized scores within day.

## 3.4 RQ2: Baseline profiles and the momentary experience of pain

Between persons, the baseline threat-value profile was reliably related to the momentary experience of pain (Figure 5, Table 5). Participants with a stronger threat-value profile attended more to pain (zero-order $r = .39$, $p = .001$; partial $r = .39$, $p = .001$ controlling mean pain) and engaged less with their activities ($r = -.30$, $p = .01$; partial $r = -.30$, $p = .01$). Both associations were specific to the threat-value profile: the biomedical profile was unrelated to attention ($r = .12$, $p = .32$) or engagement ($r = -.05$, $p = .70$), and the personality profile was weaker and, for attention, reduced after controlling pain ($r = .34$ to partial $r = .26$). Adding the three profiles to mean pain doubled the explained variance in mean attention (from $R^2 = .18$ to $R^2 = .37$) and explained variance in mean engagement that pain alone did not ($R^2 = .00$ to $R^2 = .10$). Baseline profiles did not, however, moderate the *within-person* coupling strengths: across the five pre-specified couplings and the three profiles, only one of 15 cross-level interactions reached significance, no more than expected by chance (Figure 5, Panel D, Table S10). The same pattern extended to flow: a stronger baseline threat-value profile went with a lower average flow experience ($r = -.27$, $p = .03$), as did higher neuroticism ($r = -.26$, $p = .04$), both surviving adjustment for mean pain, whereas no profile predicted the person-specific flow-pain coupling. Trait vulnerability therefore shaped how much, on average, a patient attended to pain and disengaged from activity, but not the moment-to-moment coupling strengths, which remained person-specific. H5 was thus supported at the level of mean states but not at the level of the within-person dynamics.

**Figure 5.** Baseline trait profiles and the momentary experience of pain.




*Note.* Panel A shows the partial correlation (controlling mean momentary pain) of each baseline profile with mean momentary attention to pain and mean goal engagement. Panels B and C show the threat-value profile against mean attention to pain and mean goal engagement, with an ordinary least squares fit and a 95% confidence band. Panel D shows the incremental variance in each mean state explained by adding the baseline profiles to mean pain. Panel E shows the zero-order correlations of the three profiles with the momentary states. Panel F shows the cross-level moderation of the within-person couplings by the threat-value profile, with 95% confidence intervals.

## 3.5 Robustness

The benchmark temporal circuit was stable across sensitivity analyses (Figure S7, Table S9). The 16 temporal coefficients reproduced under within-person detrending ($r = .997$), the two-item attention composite ($r = .987$), a stricter compliance threshold ($r = 1.000$), and the six-node extension, with no sign changes. In leave-one-participant-out refits, the circuit edges retained their sign in every refit. Decomposing the threat node into its two components in a five-node model confirmed that both catastrophizing and fear independently predicted a stronger subsequent focus on pain (catastrophizing $b = .087$, fear $b = .049$) and higher subsequent pain, justifying their combination into a single threat node. In the

**Table 5.** RQ2 between-person associations of the baseline accounts with mean momentary attention to pain and goal engagement.

| Mean momentary state | Account | $n$ | $r$ ($p$) | Partial $r$ ($p$) |
|---|---|---|---|---|
| Attention to pain | Threat | 67 | 0.39 (.001) | 0.39 (.001) |
| Attention to pain | Biomedical | 67 | 0.12 (.316) | -0.15 (.215) |
| Attention to pain | Personality | 64 | 0.34 (.006) | 0.26 (.040) |
| Goal engagement | Threat | 67 | -0.30 (.013) | -0.30 (.014) |
| Goal engagement | Biomedical | 67 | -0.05 (.695) | -0.05 (.695) |
| Goal engagement | Personality | 64 | -0.25 (.048) | -0.25 (.047) |

*Note.* Zero-order and partial (controlling mean momentary pain) correlations of each person's mean momentary state with the threat, biomedical, and personality accounts.

person-level bootstrap, all of the circuit edges had 95% intervals that excluded zero, including the pain-to-engagement edge ($b = -.048$, bootstrap 95% CI $[-.100, -.003]$) that had only reached a trend in the pooled Wald test, so the interruption of goal engagement by pain was supported once sampling variability in persons was taken into account (Table S8). Residual diagnostics showed near-zero within-person lag-1 residual autocorrelation. Daily functioning outcomes showed only weak within-person day-level coupling with the momentary measures (Figure S8, Table S11), consistent with the small between-day variance of pain and reinforcing that the action in these data is at the moment-to-moment level where the idiographic networks operate. Momentary measures showed a small drift across the two study weeks but no change in the temporal structure (Figure S9, Table S12).

# 4 Discussion

This study used a chronic-pain experience-sampling dataset with an idiographic lens and asked what drives momentary pain. Three findings stand out. First, at the level of the individual patient, threat and attention, rather than pain intensity alone, precede momentary pain, in a compact threat-attention-pain circuit (threat to attention $b = .125$; threat to pain $b = .096$; pain to attention $b = .083$). Second, this shared circuit summarizes wide person-to-person variation: the threat-to-pain coupling ranged from clearly negative to strongly positive, and was individually reliable in only a minority of patients. Third, a baseline threat-value profile predicted how much a patient attended to pain and disengaged from activity ($r = .39$ and $r = -.30$), but not the strength of the within-person couplings. Each research question is discussed in turn, followed by limitations and future directions.

## 4.1 Main findings

### 4.1.1 A shared but person-specific threat-attention-pain circuit (RQ1)

The momentary dynamics matched the cognitive-affective model of the interruptive function of pain (Eccleston and Crombez, 1999). The largest cross-lagged effect ran from threat to attention, the momentary expression of the model's central claim that the threat value of pain, and not its intensity alone, governs how strongly pain captures attention (Crombez et al., 1998, 2013). Threat and attention then preceded higher pain, and pain fed back into attention, closing a self-reinforcing loop that is consistent with accounts of hypervigilance and worry in chronic pain (Eccleston and Crombez, 2007; Van Damme et al., 2010; Crombez et al., 2012). The interruption of goal engagement by pain was small and reached significance only in the person-level bootstrap, which fits the observation that most everyday activities were not pain-related and that engagement is largely momentary and context-bound; it is a real but modest effect that sits at the edge of what this design can resolve. Broadening that node into the momentary flow experience did not change this picture: absorption accompanied by enjoyment tracked lower pain and, more strongly, lower pain interference at the same moment, which is what an attentional account predicts (Van Damme et al., 2010), but it predicted nothing at the next prompt. Flow is in that sense the mirror image of pain interruption rather than a cause of relief, and the daily-life evidence is therefore weaker than the flow-induced analgesia proposed on the basis of induced-flow paradigms such as virtual reality and music (Gromakovskis and Gūtmanis, 2025). Two features of everyday activity help explain the gap. Challenge in this sample co-occurred with physical exertion, which carries its own cost, and the least painful moments were not the flow moments but the low-challenge, high-skill ones, so what accompanied relief here looks more like competence without demand than optimal challenge.

The central RQ1 result, however, is that this shared circuit summarizes wide individual variation. The threat-to-pain coupling ranged from clearly negative to strongly positive across patients, and the average masked that range, a concrete instance of the group-to-individual non-generalizability that motivates within-person modelling (Fisher et al., 2018; Molenaar, 2004). When individual networks were estimated directly, they were estimable but sparse: most patients showed a strong pain autoregression and few selected cross-variable edges, and the threat-to-pain and attention-to-pain edges, when selected, were always positive. The idiographic picture is therefore a directional backbone that is broadly shared but whose magnitude, and even presence, is person-specific, which is the answer the intermediate series length here makes possible (Beltz et al., 2016; Wright and Woods, 2020).

#### 4.1.2 Trait vulnerability and the level of the momentary experience (RQ2)

At the level of mean states, the baseline threat-value profile predicted more attention to pain and less goal engagement, and did so more specifically than biomedical severity or neuroticism. This pattern supports the claim that threat appraisal, not pain severity alone, organizes the daily experience of chronic pain.

The within-person dynamics tell a different and complementary story. Baseline profiles did not moderate the coupling strengths: how strongly a patient's threat or attention predicted their next report of pain was not captured by their questionnaire scores. One reading is that trait vulnerability sets the average level of vigilance and disengagement, whereas the moment-to-moment coupling is governed by state processes and context that baseline questionnaires do not index. This dissociation between level effects, which are trait-linked, and dynamic effects, which are person-specific, is itself an argument for measuring the dynamics directly rather than inferring them from traits (Wright and Woods, 2020). Average flow behaved the same way, being lower in patients with a stronger threat profile while leaving the person-specific flow-pain coupling unpredicted.

The flow composite also carries a measurement lesson. It reproduced the benchmark network almost exactly, and its association with pain was carried by absorption and perceived skill, while enjoyment added nothing once they were included and challenge reversed sign. A construct whose components pull in opposite directions is poorly served by a summed score, and reporting the composite alone would have hidden both the driving constituent and the counteracting one (VanderWeele, 2022). The same applies to the defining challenge-skill balance, which was null within persons and negative between them, so that in this sample flow is better described by how demanding and how manageable an activity is than by how well the two are matched, an attenuation of the balance effect that is directionally consistent with meta-analytic evidence (Fong et al., 2015).

## 4.2 Limitations

The flow measure warrants its own qualification. It rests on four single items and covers three of the nine dimensions Csikszentmihalyi describes, omitting sense of control, time transformation, loss of self-consciousness, and unambiguous feedback, so it is a core proxy rather than a full measure of flow (Jackson and Eklund, 2002). Its associations with pain are concurrent, and at the roughly ninety-minute spacing of these prompts no temporal claim is available; a denser protocol may be required for a state defined by moment-to-moment absorption. The open-text activity description was not coded, so the challenge-activation confound could be adjusted for but not separated.

Several limitations qualify these conclusions. They concern, in turn, the individual estimates, the causal status of the edges, the temporal design, the measurement of the constructs, the statistical power for individual differences, and the generality of the sample.

First, although the series length supports individual estimation, the regularized individual networks were sparse, so person-specific cross-variable edges should be read as heterogeneity around a shared backbone rather than as precise individual parameters. The unregularized per-person coefficients preserve the spread but are noisy, and the mlVAR random effects shrink person-specific effects toward the group mean; the two representations bracket the true heterogeneity rather than pinpoint it.

Second, the analysis is observational. Threat, attention, and pain were not manipulated and can influence one another through unmeasured time-varying causes such as mood, medication, or activity demands, so the temporal edges are predictive within-person associations, not causal effects (Rohrer, 2018). The self-reinforcing appearance of the pain-attention loop is a statistical description, not a demonstration of a vicious cycle.

Third, the temporal resolution is coarse and unequal. Adjacent prompts are separated by variable hours, no lag is carried overnight, and momentary reports are retrospective over a short window. Some pain processes, such as the reflexive capture of attention by a pain spike, may unfold faster than this design can observe, whereas others, such as behavioural disengagement, may require longer than a single prompt interval (Hamaker and Wichers, 2017). The weak group-level interruption of engagement may reflect this mismatch of time scales as much as a weak process.

Fourth, the momentary constructs are single items or short composites, chosen to keep the diary brief. The two-item attention analysis was reassuring and the threat composite combined two appraisals, but multi-item momentary measurement would improve reliability and would allow fear and catastrophizing to be separated rather than pooled into a single threat node.

Fifth, the RQ2 sample of 67 participants with baseline data is modest for individual-difference analyses. The between-person associations were robust, but the cross-level moderation was better powered for the average coupling than for its moderation, so the absence of trait moderation should be read as a failure to detect rather than as evidence of no moderation.

Sixth, all data come from a single chronic-pain cohort recruited at one pain clinic, with a predominance of women and of back and widespread pain. Replication in more diverse samples, and in other pain conditions, is needed before the threat-attention-pain circuit is generalized.

### 4.3 Future directions

Four directions follow from these findings. First, and most directly, single-case experimental and N-of-1 designs are needed to move from prediction toward mechanism. Because the threat-to-attention and threat-to-pain couplings are the empirical core of the circuit, they are natural intervention targets: a within-person experiment that reduces threat appraisals, for example through reappraisal or exposure, could test whether a patient's own threat-to-pain coupling weakens, which prediction alone cannot establish (Wright and Woods, 2020; Crombez et al., 2012).

Second, the design should be enriched to resolve the time scales at which these processes operate. Denser sampling around pain episodes, event-contingent prompts triggered by pain spikes, and the addition of passive context sensing would let the fast, reflexive capture of attention be separated from slower behavioural disengagement, and would give individual networks the series length needed to estimate cross-variable edges with more precision (Beltz et al., 2016).

Third, the node set should be expanded and refined. Separating fear from catastrophizing, adding additional goal-directed activity characteristics measured in the diary, and including mood and medication as covariates would test whether the threat-attention-pain circuit is specific or embedded in a broader affective-motivational system, and whether goal engagement interrupts pain more clearly when it is measured as sustained absorption rather than momentary state.

Fourth, because trait vulnerability predicted the level but not the dynamics of the momentary experience, person-specific coupling estimates may carry information beyond questionnaires. Whether these estimates can be used, alongside baseline profiles, to tailor attention- and threat-focused interventions is an open and clinically consequential question that larger and longer studies should address directly.

## 5 Conclusion

This idiographic network study of a chronic-pain diary showed that momentary pain, its threat value, attention to pain, and goal engagement form a within-person threat-attention-pain circuit in which threat and attention, rather than pain intensity alone, precede momentary pain. The circuit is shared in direction but strongly person-specific in magnitude. A baseline threat-value profile predicted how much a patient attended to pain and disengaged from activity, but it did not predict the strength of the within-person couplings. The findings support measuring the dynamics of threat and attention directly in daily life, and using person-specific estimates, rather than baseline traits alone, as a basis for understanding and eventually tailoring psychological pain care.

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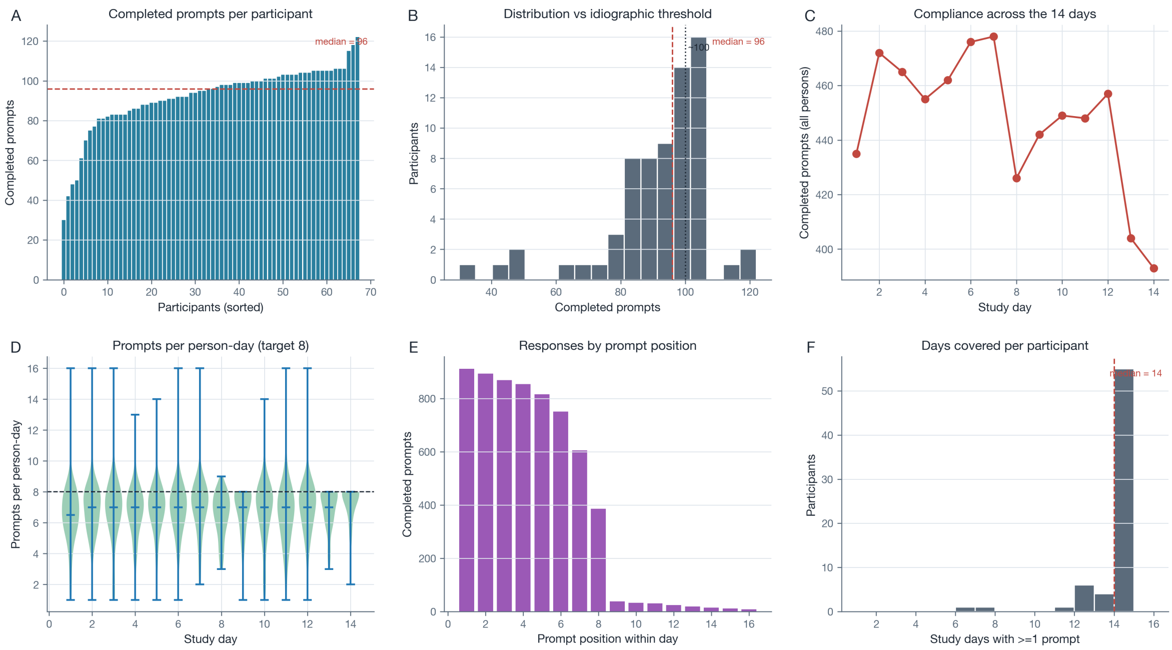
# Appendix

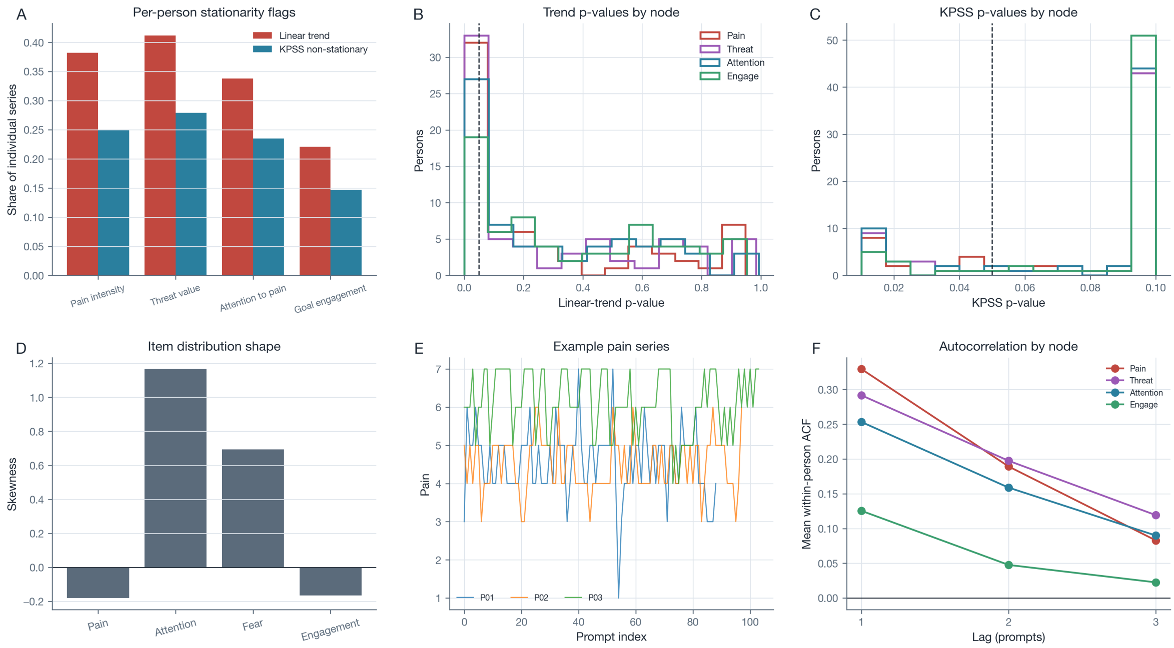
## Supplementary Figures

**Figure S1.** Design, compliance, and data quality.



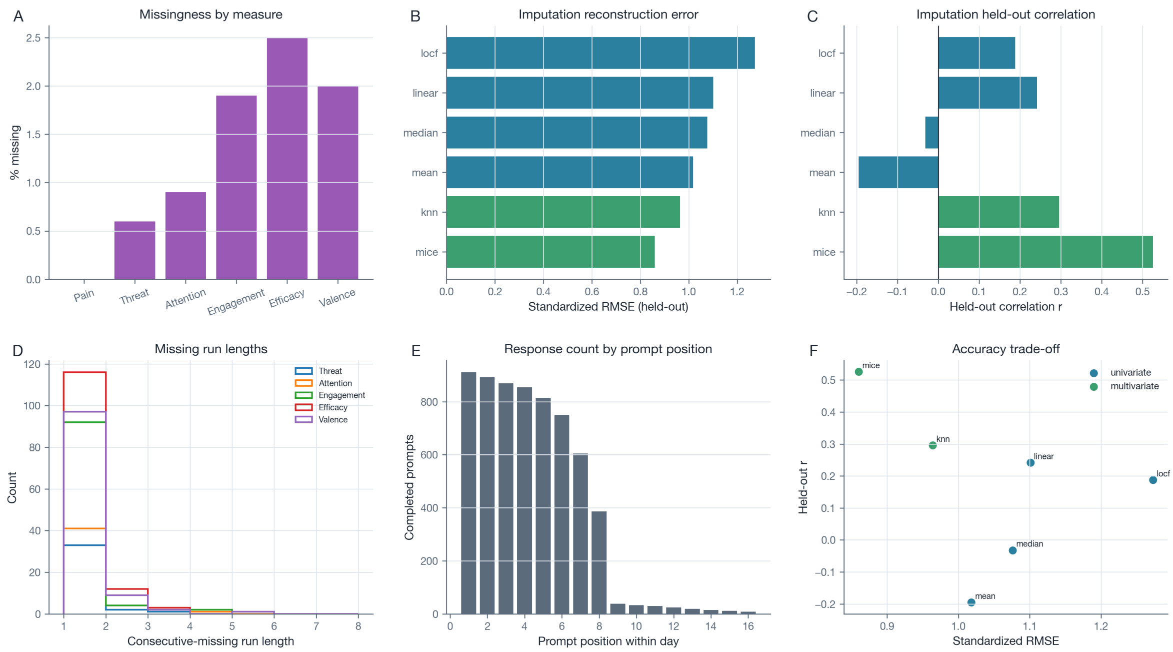

*Note.* Panel A shows completed prompts per participant with the median. Panel B shows the distribution of completed prompts against the idiographic threshold. Panel C shows compliance across the 14 days. Panel D shows prompts per person-day (violins) against the target of eight. Panel E shows responses by prompt position within the day. Panel F shows study days covered per participant.

**Figure S2.** Item distributions and per-person stationarity.



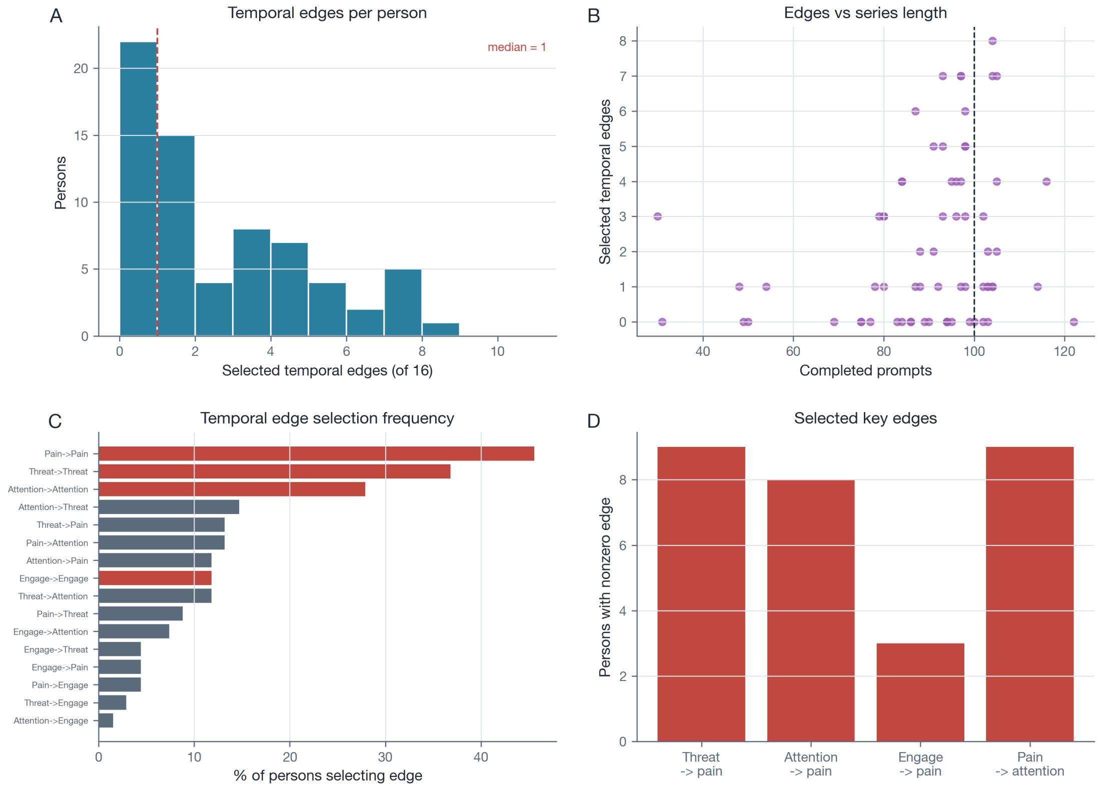

*Note.* Panel A shows the share of individual series flagged by a linear-trend test and by KPSS for each core node. Panels B and C show the distributions of per-person linear-trend and KPSS p-values. Panel D shows the skewness of the core items. Panel E shows example within-person pain series. Panel F shows the mean within-person autocorrelation of each node at lags one to three.

**Figure S3.** Missingness and imputation benchmark.



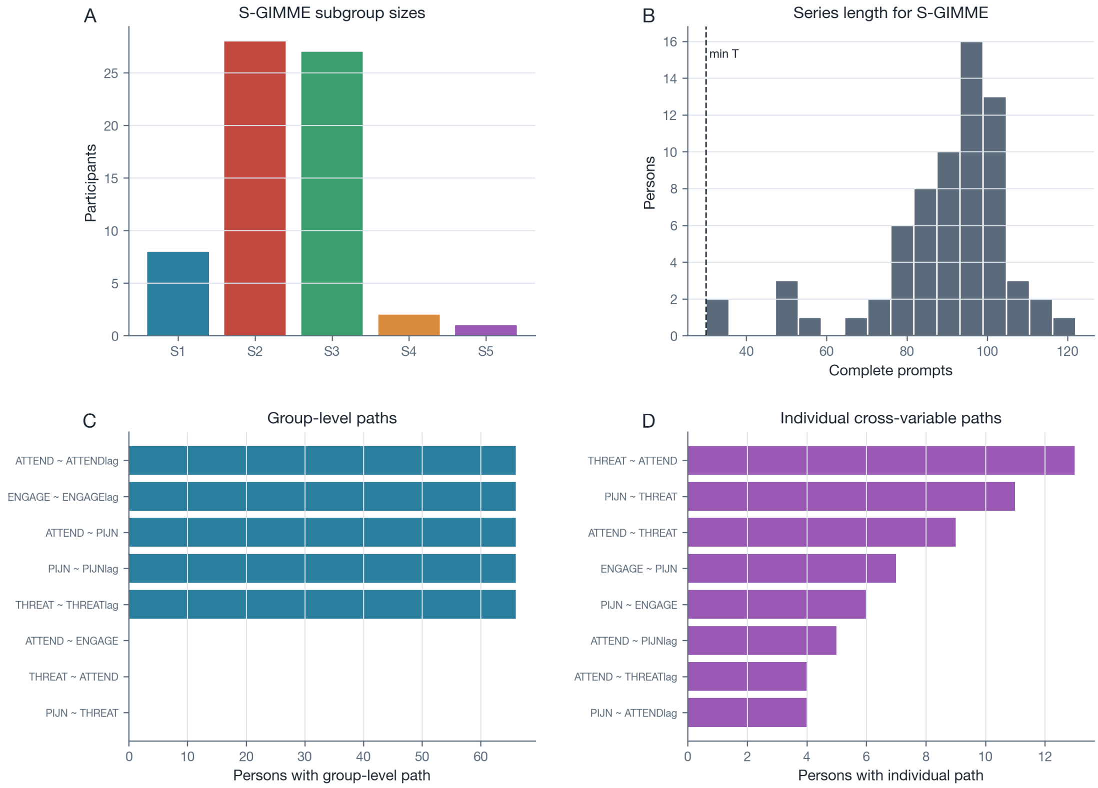

*Note.* Panel A shows momentary missingness by measure. Panels B and C show the held-out reconstruction error and correlation of univariate and multivariate imputation methods. Panel D shows the distribution of consecutive-missing run lengths. Panel E shows response counts by prompt position. Panel F shows the accuracy trade-off between reconstruction error and held-out correlation.

**Figure S4.** Per-person graphicalVAR feasibility.



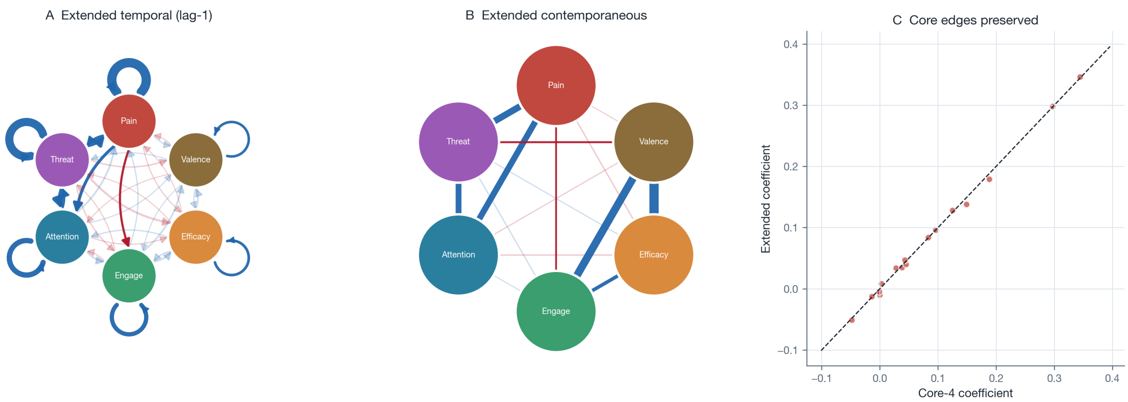

*Note.* Panel A shows selected temporal edges per person with the median. Panel B plots selected edges against series length. Panel C shows the temporal edge-selection frequency across persons. Panel D shows the number of participants with a nonzero key edge.

**Figure S5.** S-GIMME shared and individual structure.



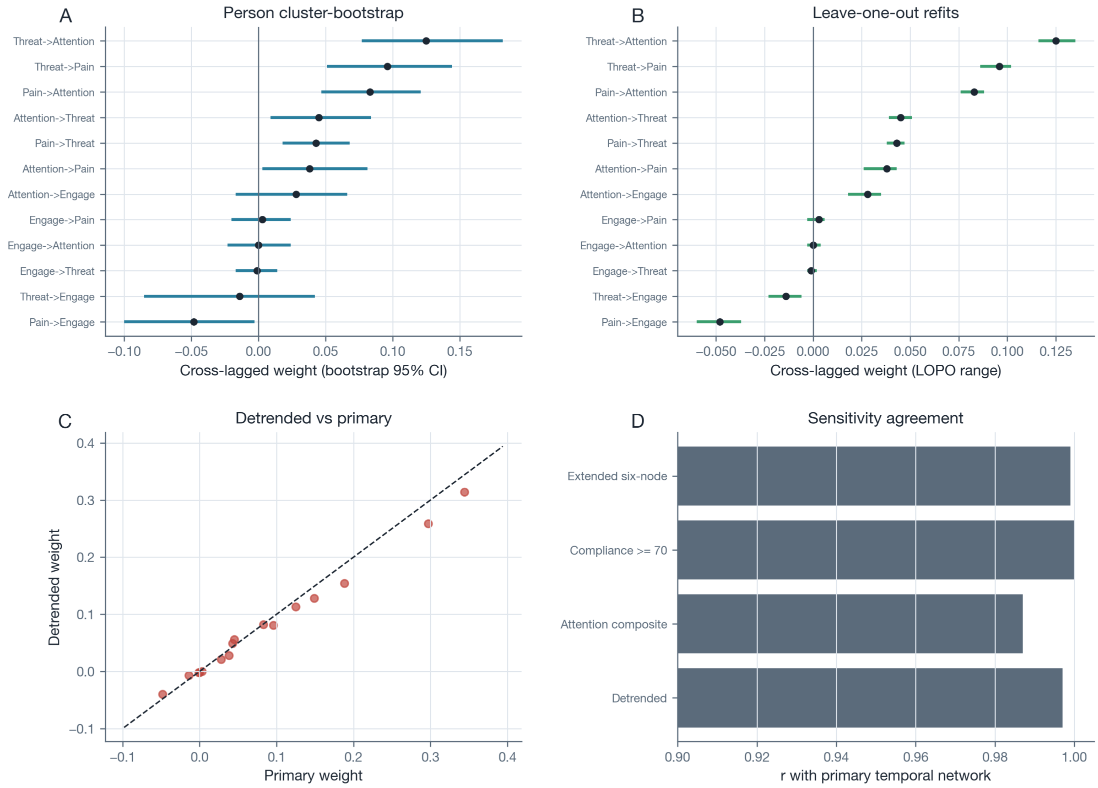

*Note.* Panel A shows the S-GIMME subgroup sizes. Panel B shows the series length available for S-GIMME. Panel C lists the most common S-GIMME group-level paths, with counts indicating the participants for whom each path was retained at the group level. Panel D lists the most common individual-level cross-variable paths.

**Figure S6.** Extended six-node model.



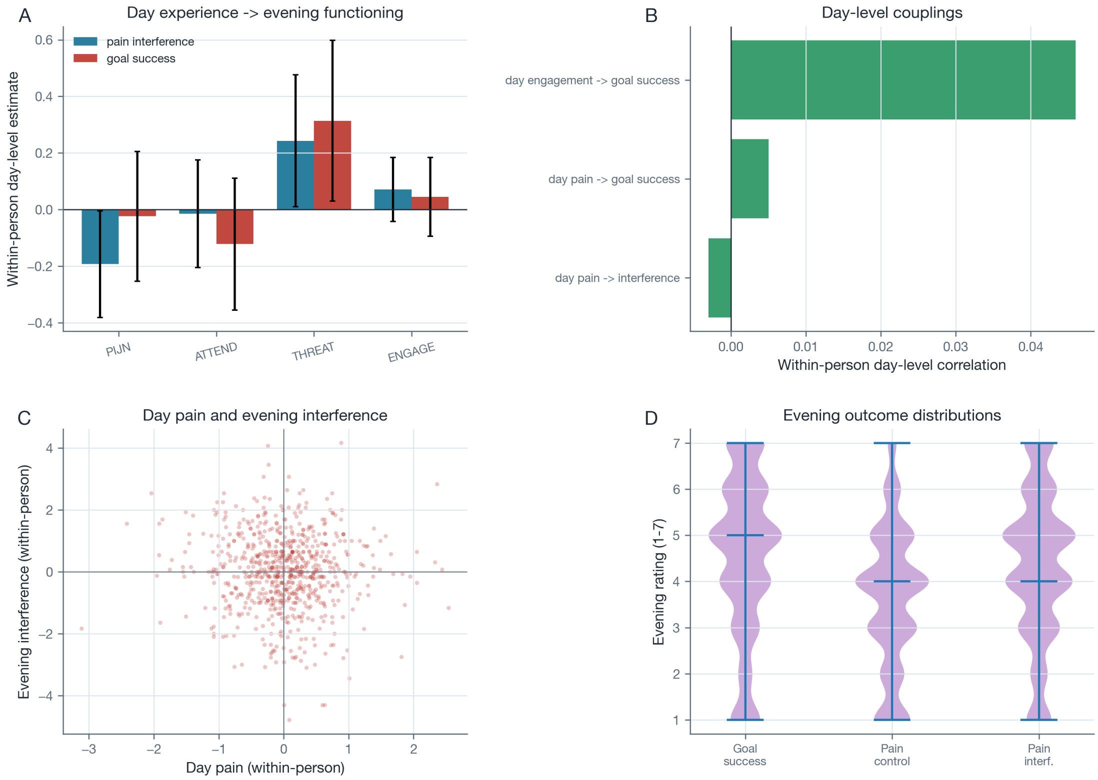

*Note.* Panels A and B show the temporal and contemporaneous networks of the six-node model, which adds momentary efficacy and valence. Panel C compares the core four-node temporal coefficients between the primary and extended models.

**Figure S7.** Robustness of the temporal circuit.



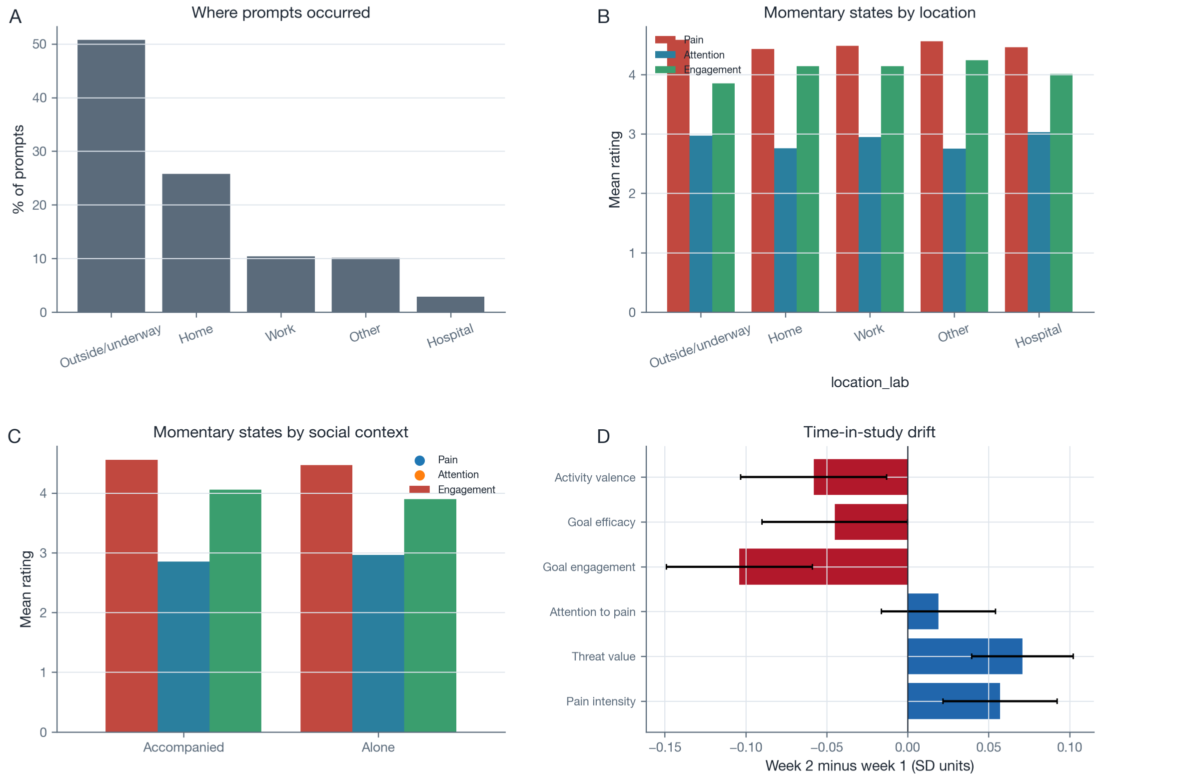

*Note.* Panel A shows person cluster-bootstrap 95% intervals for the cross-lagged edges. Panel B shows leave-one-participant-out ranges. Panel C compares within-person detrended coefficients with the primary coefficients. Panel D shows the edge-weight correlation of each sensitivity refit with the primary network.

**Figure S8.** Daily dysfunction dynamics.



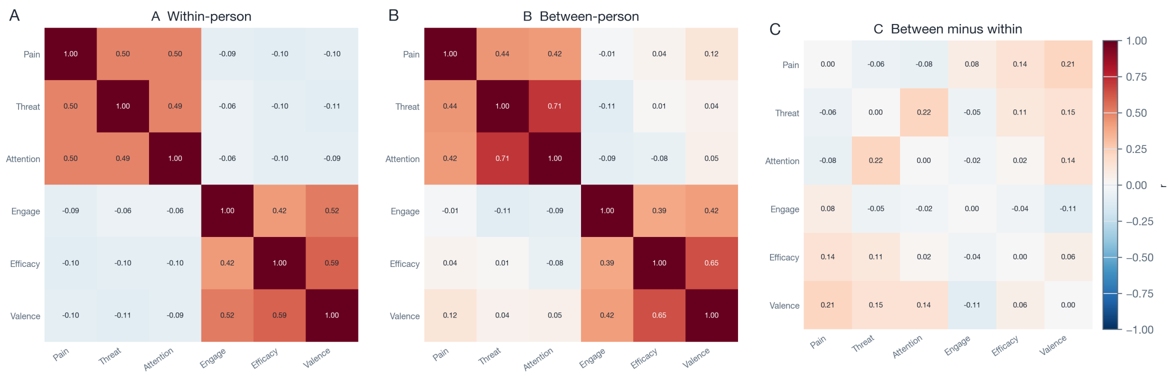

*Note.* Panel A shows within-person day-level models relating the day's momentary experience to the evening reports of pain interference and goal success. Panel B shows within-person day-level correlations. Panel C shows day pain against evening interference within persons. Panel D shows the distributions of the evening outcomes.

**Figure S9.** Everyday context and time-in-study.



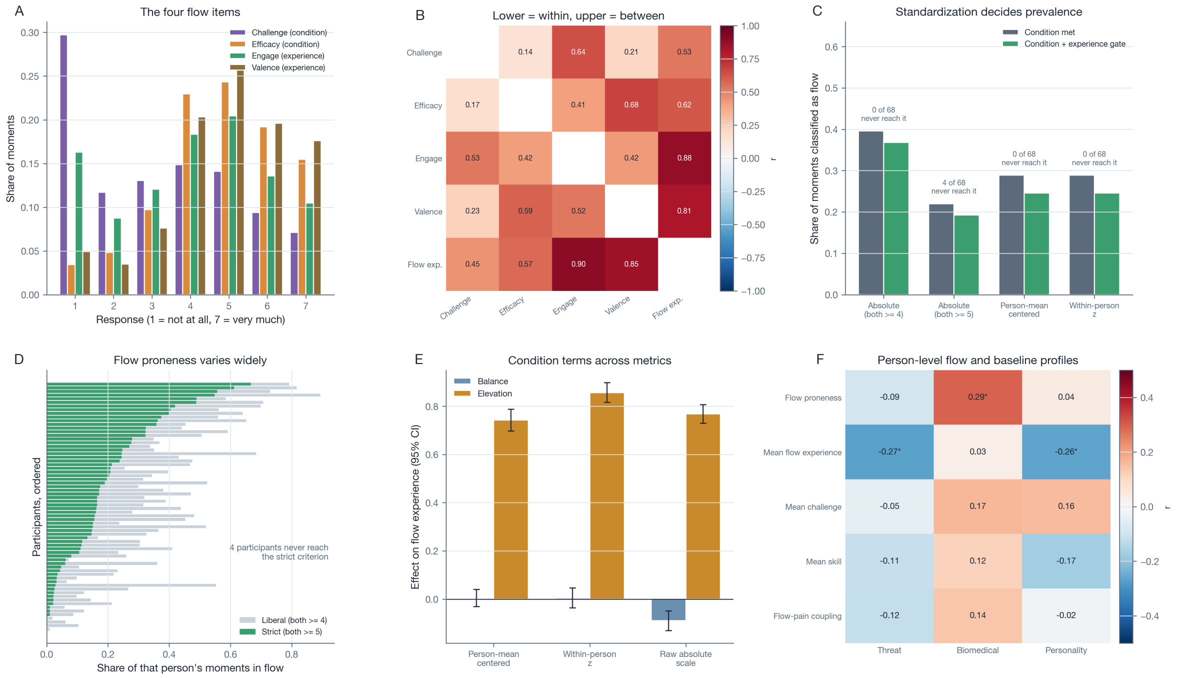

*Note.* Panel A shows where prompts occurred. Panel B shows the momentary states by location. Panel C shows the momentary states when alone versus accompanied. Panel D shows the week-2 minus week-1 drift for each measure with 95% confidence intervals from participant-clustered mixed models.

**Figure S10.** Within-person and between-person correlations.




*Note.* Panel A shows correlations after within-person centring. Panel B shows correlations among person means. Panel C shows the between-person minus within-person difference matrix.

**Figure S11.** Flow operationalization and standardization.




*Note.* Panel A shows the response distribution of the four activity appraisals, labelled by their role in the flow model. Panel B shows their correlation structure, within persons below the diagonal and between persons above it. Panel C shows the share of moments classified as flow under each standardization rule, before and after the experience gate, annotated with the number of participants who never reach the criterion; only the absolute rules allow a participant to score zero. Panel D shows the per-person share of flow moments under the liberal and the strict absolute criterion. Panel E shows the balance and elevation estimates on each metric. Panel F shows the person-level flow variables against the three baseline profiles, with an asterisk marking $p < .05$.

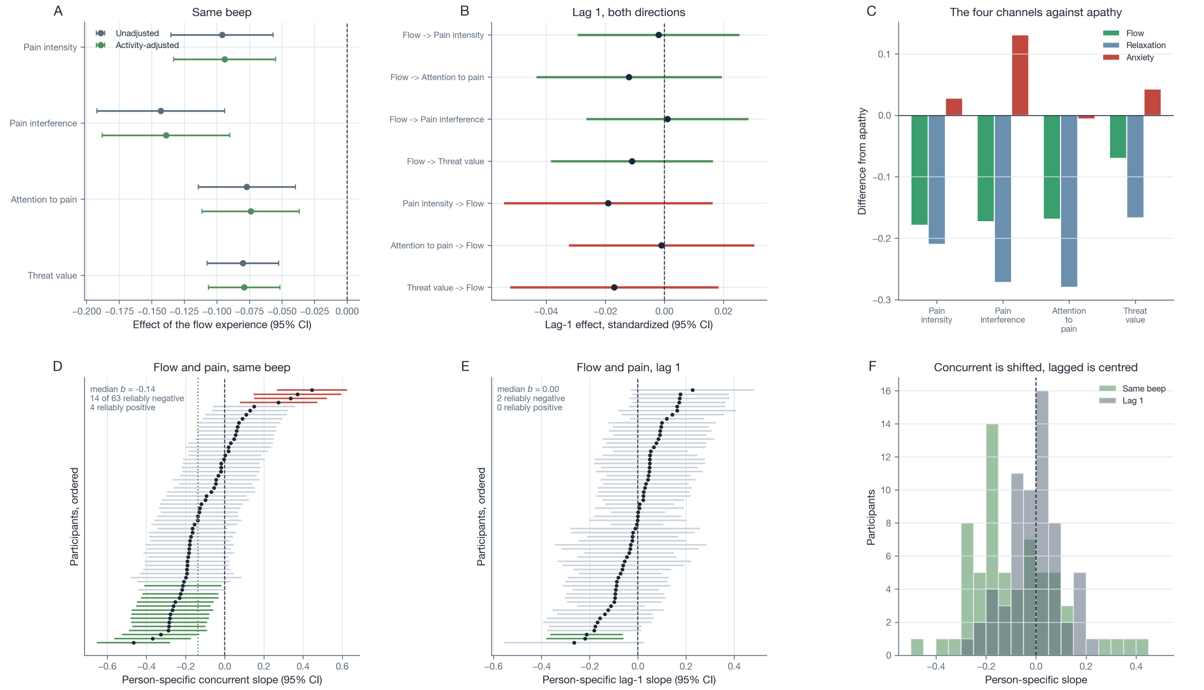
## Supplementary Tables

**Table S1.** Contemporaneous and between-person networks (benchmark).

| Network | Edge | Partial $r$ | $p$ |
|---|---|---|---|
| Contemporaneous | Threat value - Pain intensity | 0.327 | <.001 |
| Contemporaneous | Attention to pain - Pain intensity | 0.282 | <.001 |
| Contemporaneous | Attention to pain - Threat value | 0.271 | <.001 |
| Contemporaneous | Goal engagement - Pain intensity | -0.064 | .003 |
| Contemporaneous | Goal engagement - Threat value | -0.01 | .462 |
| Contemporaneous | Goal engagement - Attention to pain | -0.022 | .188 |
| Between-person | Threat value - Pain intensity | 0.227 | .051 |
| Between-person | Attention to pain - Pain intensity | 0.177 | .119 |
| Between-person | Attention to pain - Threat value | 0.642 | <.001 |
| Between-person | Goal engagement - Pain intensity | 0.051 | .598 |
| Between-person | Goal engagement - Threat value | -0.097 | .362 |
| Between-person | Goal engagement - Attention to pain | -0.024 | .791 |

*Note.* Partial correlations from the pooled mlVAR benchmark.

**Figure S12.** Flow and the momentary experience of pain.




*Note.* Panel A shows the concurrent association of the flow experience with the four momentary pain measures, unadjusted and adjusted for physical activation. Panel B shows all lag-1 effects in both directions, on person-standardized scores within day. Panel C shows the four challenge-skill channels against apathy as the reference. Panels D and E show the person-specific concurrent and lag-1 flow-pain slopes with 95% confidence intervals, ordered by size and coloured by whether the interval excludes zero. Panel F overlays the two person-level distributions.

**Table S2.** Item distribution and per-person stationarity.

*Panel A.* Skewness and kurtosis of the core momentary items.

| Measure | Skewness | Kurtosis |
|---|---|---|
| Pain | -0.18 | -0.528 |
| Attention | 1.167 | 0.776 |
| Fear | 0.695 | -0.126 |
| Engagement | -0.165 | -1.055 |

*Panel B.* Share of individual series flagged by a linear-trend test and by KPSS.

| Variable | Prop. linear trend | Prop. KPSS non-stationary |
|---|---|---|
| Pain intensity | 0.382 | 0.25 |
| Threat value | 0.412 | 0.279 |
| Attention to pain | 0.338 | 0.235 |
| Goal engagement | 0.221 | 0.147 |

*Note.* KPSS = Kwiatkowski-Phillips-Schmidt-Shin test (null is stationarity). A within-person detrended sensitivity model is reported in the robustness battery.

**Table S3.** Missing data and imputation benchmark.

*Panel A.* Momentary missingness by measure.

| Measure | Observed | Missing | % missing |
|---|---|---|---|
| Pain | 6262 | 0 | 0.0 |
| Threat | 6222 | 40 | 0.6 |
| Attention | 6203 | 59 | 0.9 |
| Engagement | 6143 | 119 | 1.9 |
| Efficacy | 6108 | 154 | 2.5 |
| Valence | 6136 | 126 | 2.0 |

*Panel B.* Univariate and multivariate imputation compared by held-out reconstruction.

| Method | Type | Std. RMSE | Std. MAE | *r* (held-out) |
|---|---|---|---|---|
| mice | multivariate | 0.86 | 0.653 | 0.526 |
| knn | multivariate | 0.964 | 0.741 | 0.296 |
| mean | univariate | 1.018 | 0.804 | -0.195 |
| median | univariate | 1.076 | 0.748 | -0.032 |
| linear | univariate | 1.101 | 0.803 | 0.242 |
| locf | univariate | 1.273 | 0.875 | 0.188 |

*Note.* The primary temporal models use available-case (pairwise) handling within each equation. As a benchmark, 20% of observed values were held out at random and reconstructed. Multivariate methods that borrow strength across concurrent measures recovered momentary fluctuations best, which supports the available-case decision.

**Table S4.** S-GIMME path prevalence.

| Path | Class | Group *n* | Individual *n* |
|---|---|---|---|
| ATTEND ~ ATTENDlag | Autoregressive | 66 | 0 |
| ENGAGE ~ ENGAGElag | Autoregressive | 66 | 0 |
| ATTEND ~ PIJN | Cross-variable | 66 | 0 |
| PIJN ~ PIJNlag | Autoregressive | 66 | 0 |
| THREAT ~ THREATlag | Autoregressive | 66 | 0 |
| THREAT ~ ATTEND | Cross-variable | 0 | 13 |
| PIJN ~ THREAT | Cross-variable | 0 | 11 |
| ATTEND ~ THREAT | Cross-variable | 0 | 9 |
| ENGAGE ~ PIJN | Cross-variable | 0 | 7 |
| PIJN ~ ENGAGE | Cross-variable | 0 | 6 |
| ATTEND ~ PIJNlag | Cross-variable | 0 | 5 |
| PIJN ~ ATTENDlag | Cross-variable | 0 | 4 |
| ATTEND ~ ENGAGE | Cross-variable | 0 | 4 |
| THREAT ~ PIJN | Cross-variable | 0 | 4 |

*Note.* Counts are the number of participants for whom S-GIMME retained each path at the group or individual level. Paths place the outcome on the left of the tilde; 'lag' marks lagged (temporal) predictors.

**Table S5.** Per-person graphicalVAR: temporal edge-selection frequency.

| Directed edge | Type | % of persons selecting |
|---|---|---|
| Pain intensity → Pain intensity | Autoregressive | 45.6 |
| Threat value → Threat value | Autoregressive | 36.8 |
| Attention to pain → Attention to pain | Autoregressive | 27.9 |
| Attention to pain → Threat value | Cross-variable | 14.7 |
| Pain intensity → Attention to pain | Cross-variable | 13.2 |
| Threat value → Pain intensity | Cross-variable | 13.2 |
| Threat value → Attention to pain | Cross-variable | 11.8 |
| Goal engagement → Goal engagement | Autoregressive | 11.8 |
| Attention to pain → Pain intensity | Cross-variable | 11.8 |
| Pain intensity → Threat value | Cross-variable | 8.8 |
| Goal engagement → Attention to pain | Cross-variable | 7.4 |
| Pain intensity → Goal engagement | Cross-variable | 4.4 |
| Goal engagement → Pain intensity | Cross-variable | 4.4 |
| Goal engagement → Threat value | Cross-variable | 4.4 |
| Threat value → Goal engagement | Cross-variable | 2.9 |
| Attention to pain → Goal engagement | Cross-variable | 1.5 |

*Note.* Percentage of fitted participants for whom the regularized (EBIC) individual network selected each directed lag-1 edge. Autoregressive edges dominate; cross-variable edges are sparse and heterogeneous.

**Table S6.** Extended six-node model: autoregressive effects and effects on pain.

| Effect (lag-1) | Estimate | SE | *p* | Random SD |
|---|---|---|---|---|
| Threat value → Pain intensity | 0.096 | 0.027 | <.001 | 0.137 |
| Attention to pain → Pain intensity | 0.035 | 0.019 | .064 | 0.069 |
| Goal engagement → Pain intensity | 0.008 | 0.013 | .544 | 0.037 |
| Activity valence → Pain intensity | -0.005 | 0.014 | .736 | 0.041 |
| Goal efficacy → Pain intensity | -0.003 | 0.013 | .817 | 0.0 |
| Pain intensity → Pain intensity | 0.346 | 0.025 | <.001 | 0.153 |
| Threat value → Threat value | 0.298 | 0.028 | <.001 | 0.171 |
| Attention to pain → Attention to pain | 0.179 | 0.026 | <.001 | 0.156 |
| Goal engagement → Goal engagement | 0.138 | 0.024 | <.001 | 0.128 |
| Goal efficacy → Goal efficacy | 0.07 | 0.018 | <.001 | 0.027 |
| Activity valence → Activity valence | 0.065 | 0.019 | .001 | 0.021 |

*Note.* The model adds two further goal-directed activity characteristics, momentary efficacy and valence. The core four-node edges were essentially unchanged relative to the primary benchmark (see the sensitivity-agreement table).

**Table S7.** Per-person unregularized pain-driver couplings.

| Driver → pain | *n* | Mean | SD | Min | Max | Indiv. sig. |
|---|---|---|---|---|---|---|
| Threat value | 66 | 0.082 | 0.204 | -0.24 | 0.86 | 11 |
| Attention to pain | 66 | 0.068 | 0.174 | -0.31 | 0.73 | 7 |
| Goal engagement | 66 | 0.001 | 0.133 | -0.53 | 0.31 | 5 |

*Note.* Each participant's lag-1 coefficient predicting pain from the previous prompt's driver, on person-standardized data. Indiv. sig. counts participants whose 95% confidence interval excludes zero.

**Table S8.** Stability of pooled cross-lagged edges: leave-one-out and person bootstrap.

| Cross-lagged effect | Mean | LOPO range | Bootstrap 95% CI | Bootstrap sign-consistency |
|---|---|---|---|---|
| Pain intensity → Goal engagement | -0.048 | [-0.060, -0.037] | [-0.100, -0.003] | 0.98 |
| Threat value → Goal engagement | -0.014 | [-0.023, -0.006] | [-0.085, 0.042] | 0.7 |
| Goal engagement → Threat value | -0.001 | [-0.003, 0.002] | [-0.017, 0.014] | 0.52 |
| Goal engagement → Attention to pain | -0.0 | [-0.003, 0.004] | [-0.023, 0.024] | 0.015 |
| Goal engagement → Pain intensity | 0.003 | [-0.003, 0.006] | [-0.020, 0.024] | 0.545 |
| Attention to pain → Goal engagement | 0.028 | [0.018, 0.035] | [-0.017, 0.066] | 0.835 |
| Attention to pain → Pain intensity | 0.038 | [0.026, 0.043] | [0.003, 0.081] | 0.98 |
| Pain intensity → Threat value | 0.043 | [0.038, 0.047] | [0.018, 0.068] | 1.0 |
| Attention to pain → Threat value | 0.045 | [0.039, 0.051] | [0.009, 0.084] | 0.995 |
| Pain intensity → Attention to pain | 0.083 | [0.076, 0.088] | [0.047, 0.121] | 1.0 |
| Threat value → Pain intensity | 0.096 | [0.086, 0.102] | [0.051, 0.144] | 1.0 |
| Threat value → Attention to pain | 0.125 | [0.116, 0.135] | [0.077, 0.182] | 1.0 |

*Note.* LOPO = leave-one-participant-out refits; bootstrap = person cluster resamples.

**Table S9.** Agreement of each sensitivity analysis with the primary temporal network.

| Sensitivity analysis | $r$ with primary | Sign flips |
|---|---|---|
| Within-person detrended | 0.997 | 0 |
| Attention two-item composite | 0.987 | 0 |
| Stricter compliance (≥ 70) | 1.0 | 0 |
| Extended six-node model | 0.999 | 0 |

*Note.* $r$ = correlation of the 16 temporal coefficients with the primary benchmark.

**Table S10.** Cross-level moderation of the within-person couplings by baseline accounts.

| Coupling | Moderator | Within-person $b$ | Interaction $b$ | Interaction SE | Interaction $p$ |
|---|---|---|---|---|---|
| Attention → Pain | Threat | 0.044 | -0.02 | 0.019 | .297 |
| Attention → Pain | Biomedical | 0.045 | -0.04 | 0.018 | .031 |
| Attention → Pain | Personality | 0.046 | 0.002 | 0.02 | .935 |
| Threat → Pain | Threat | 0.055 | -0.007 | 0.019 | .728 |
| Threat → Pain | Biomedical | 0.055 | -0.019 | 0.018 | .297 |
| Threat → Pain | Personality | 0.046 | 0.019 | 0.018 | .289 |
| Pain → Attention | Threat | 0.087 | -0.03 | 0.02 | .141 |
| Pain → Attention | Biomedical | 0.088 | -0.037 | 0.02 | .068 |
| Pain → Attention | Personality | 0.088 | -0.016 | 0.021 | .452 |
| Engagement → Attention | Threat | 0.006 | 0.004 | 0.016 | .784 |
| Engagement → Attention | Biomedical | 0.006 | -0.003 | 0.016 | .843 |
| Engagement → Attention | Personality | -0.002 | 0.015 | 0.016 | .345 |
| Pain → Engagement | Threat | -0.026 | -0.005 | 0.018 | .764 |
| Pain → Engagement | Biomedical | -0.025 | -0.018 | 0.017 | .286 |
| Pain → Engagement | Personality | -0.025 | -0.003 | 0.018 | .883 |

*Note.* Each row is a multilevel model in which a baseline account moderates a within-person lag-1 coupling (random slope). Within-person $b$ is the average coupling; interaction $b$ is the moderation by the standardized account. Couplings were not reliably moderated.

**Table S11.** Daily dysfunction: within-person day-level models.

| Evening outcome | Day predictor (within-person) | Estimate | SE | *p* |
|---|---|---|---|---|
| pain interference | Pain intensity | -0.193 | 0.096 | .045 |
| pain interference | Attention to pain | -0.015 | 0.097 | .878 |
| pain interference | Threat value | 0.243 | 0.119 | .041 |
| pain interference | Goal engagement | 0.071 | 0.058 | .228 |
| goal success | Pain intensity | -0.024 | 0.117 | .840 |
| goal success | Attention to pain | -0.122 | 0.119 | .307 |
| goal success | Threat value | 0.314 | 0.145 | .031 |
| goal success | Goal engagement | 0.045 | 0.071 | .525 |

*Note.* Multilevel models with participant random intercepts. Predictors are person-centered day means. Day-to-day (between-day) variance in pain was small, so day-level effects are weak relative to the momentary dynamics.

**Table S12.** Everyday context and time-in-study.

*Panel A.* Momentary measures by location.

| Location | *n* | % | Pain | Attention | Engagement |
|---|---|---|---|---|---|
| Outside/underway | 3172 | 50.73 | 4.58 | 2.97 | 3.85 |
| Home | 1612 | 25.78 | 4.43 | 2.76 | 4.14 |
| Work | 652 | 10.43 | 4.48 | 2.95 | 4.14 |
| Other | 636 | 10.17 | 4.56 | 2.75 | 4.24 |
| Hospital | 181 | 2.89 | 4.46 | 3.03 | 4.01 |

*Panel B.* Momentary measures by social context.

| Social context | *n* | Pain | Attention | Engagement |
|---|---|---|---|---|
| Accompanied | 4073 | 4.56 | 2.85 | 4.05 |
| Alone | 2142 | 4.47 | 2.97 | 3.9 |

*Panel C.* Week-2 versus week-1 drift (standardized, participant-clustered mixed models).

| Measure | Week-2 effect | SE | *p* |
|---|---|---|---|
| Pain intensity | 0.057 | 0.018 | .002 |
| Threat value | 0.071 | 0.016 | <.001 |
| Attention to pain | 0.019 | 0.018 | .300 |
| Goal engagement | -0.104 | 0.023 | <.001 |
| Goal efficacy | -0.045 | 0.023 | .052 |
| Activity valence | -0.058 | 0.023 | .012 |

*Note.* Context summaries are descriptive. Week-2 effects test reactivity and habituation across the two-week protocol.

**Table S13.** Flow operationalization: channels, prevalence, and standardization sensitivity.

*Panel A.* Momentary profile of the four channels (absolute rule, both items ≥ 5).

| Channel | *n* | | | | | | |
|---|---|---|---|---|---|---|---|
| Flow | 1314 | 22.0 | 4.9 | 4.15 | 2.99 | 2.34 | 5.79 |
| Relaxation | 2214 | 37.0 | 4.29 | 3.52 | 2.69 | 2.09 | 4.64 |
| Anxiety | 516 | 8.6 | 4.83 | 4.28 | 3.11 | 2.23 | 4.44 |
| Apathy | 1932 | 32.3 | 4.49 | 3.94 | 3.0 | 2.31 | 3.33 |

*Panel B.* Prevalence of flow moments under each standardization rule.

| Standardization rule | Condition met ( | | | |
|---|---|---|---|---|
| A. Absolute, both items >= 4 | 39.6 | 36.8 | 35.0 | 0 |
| A. Absolute, both items >= 5 | 22.0 | 19.3 | 16.4 | 4 |
| B. Grand-mean z, above the sample mean | 30.6 | 27.8 | 26.7 | 2 |
| C. Within-person z, above own mean | 28.9 | 24.6 | 24.5 | 0 |
| D. Person-mean centered, above own mean | 28.9 | 24.6 | 24.5 | 0 |

*Panel C.* Condition terms predicting the flow experience on each metric.

| Metric | Term | *b* | SE | *p* |
|---|---|---|---|---|
| D. Person-mean centered (primary) | Balance | 0.004 | 0.018 | .826 |
| D. Person-mean centered (primary) | Elevation | 0.743 | 0.023 | <.001 |
| C. Within-person z | Balance | 0.005 | 0.021 | .806 |
| C. Within-person z | Elevation | 0.857 | 0.021 | <.001 |
| B. Grand-mean z | Balance | -0.025 | 0.025 | .315 |
| B. Grand-mean z | Elevation | 0.89 | 0.022 | <.001 |
| A. Raw absolute scale | Balance | -0.09 | 0.021 | <.001 |
| A. Raw absolute scale | Elevation | 0.768 | 0.02 | <.001 |

*Note.* The gated criterion requires both the challenge-skill condition and a high flow experience. Within-person standardization forces a relative distribution and therefore assigns above-average moments to every participant, which is why the absolute rules are reported alongside it.